

HypatiaX Submission — Independent Audit

Cross-checked: ([jmlr_paper_main.pdf](#)), ([supp_routing_improvements.pdf](#)) (Supp. A), ([supp_benchmark_report.pdf](#)) (Supp. B), ([HypatiaX_Paper_Audit_Report_Jul09.html](#))

Cover letter: not reviewed — no cover letter file was included in the uploads. The only non-PDF/HTML file provided ([submission-final.txt](#)) contains a single line of unrelated UI text ("This chat has 96 of 100 images...") and no cover-letter content. If a cover letter exists, please upload it separately.

0. First, a problem with the audit report itself

([HypatiaX_Paper_Audit_Report_Jul09.html](#)) is not independent verification — it's misleading as currently generated. Every one of its six audit notebooks (NB-01 through NB-06) reports:

! Notebook not found — the upstream job may have failed or been skipped.

Despite this, the report's summary section asserts "*All nine abstract claims pass verification ([OK] across every check in NB-04 Step 2)*" and "*Open Issues (0 — none ✓)*" with 14 items marked "RESOLVED." The footer confirms why: "*Generated by build_report.py — no notebooks were re-executed.*" This is a stale/cached "green" summary, not a result of this run's checks. **Don't cite this report as evidence the paper is clean** — none of its checks actually executed this cycle. (And as shown below, independently re-checking the source documents turns up real issues this report doesn't catch.)

1. Issues I found by cross-checking the documents directly

● Critical — Figures 3 & 4 contradict Table 6 (same equations, same benchmark)

Figures 3 and 4 in the main paper are captioned as showing the **Core-15 benchmark** — the same 15 equations reported in **Table 6**. Table 6 gives full numeric values for every equation/method/range. Comparing them:

Equation	Method	Table 6 (authoritative)	Figure 4 heatmap cell	Match?
Arrhenius	PySR-only, Near	-0.9783	0.99	✗
Arrhenius	PySR-only, Med	-0.6766	0.43	✗
Constant Product	HypatiaX, Near	0.9996	astronomically large garbled string ($\sim 10^{29}$)	✗
Michaelis-Menten	HypatiaX, Near/Med/Far	-0.0979 / -2.4717 / -634.60	(nan) / (nan) / (nan)	✗
Logistic Growth	HypatiaX, Near/Med/Far	0.9999 / 1.0000 / 0.9999	(nan) / (nan) / (nan)	✗
Value at Risk	HypatiaX, Near/Med/Far	0.9999 / 1.0000 / 0.9999	(nan) / (nan) / (nan)	✗
Gravitational Force	HypatiaX, Near/Med/Far	-2.6752 / -0.0016 / -7.6360	(nan) / (nan) / (nan)	✗

Figure 4's DeFi-AMM/Hard cell literally renders as a ~130-digit number (`(-59791519071875056500074578...)`), and Figure 3 — captioned as "clipped to $[-1, 1]$ for visual clarity" — shows the same kind of astronomical garbage value in one cell, which is impossible if clipping were actually applied.

What this means: the figure-generation pipeline is silently producing NaNs and overflow artifacts wherever R^2 is not near-perfect, and the paper is shipping these broken figures alongside a table (Table 6) that has the correct numbers. A reviewer who only looks at the figures would see "undefined" or nonsensical results exactly where the paper's most important findings live (Michaelis-Menten's catastrophic collapse, Arrhenius's scale-incompatibility failure) — undercutting the very point the figures are supposed to illustrate. **These figures need to be regenerated from the Table 6 JSON before submission**, not just relabeled.

● Critical — Section 10.7's "reduction" claim is numerically backwards

"The reduction from the random-split baseline (**0.678**) to the PCA-split corrected result (**0.789**) is expected: the PCA-directed split is strictly harder..."

Two problems:

1. **0.789 > 0.678**. Going from 0.678 to 0.789 is an *increase*, not a "reduction" — and the text's own reasoning (PCA split is harder → should produce a *lower* score) requires the

opposite direction.

2. **0.678 doesn't match the number it's supposed to summarize.** The "primary result" it refers to is $9/30 = 0.300$ (this exact $9/30$ figure is what's reported in the abstract, Table 7, and Section 10.7's own opening sentence). 0.678 doesn't correspond to any number derivable from $9/30$ or from $142/180$ (0.789).

This paragraph is self-contradictory as written and will draw a referee's attention immediately. It needs a rewrite with the actual, verified numbers plugged in — right now it reads as an unedited placeholder or a copy-paste from a different comparison.

● Moderate — Supp. B's own headline number for HLLMNN (M4) disagrees with itself

- **Table 4** (six-method comparison, "noiseless protocol, $R^2 \geq 0.9999$ "): HLLMNN = **30/30 (100%)**, footnoted "v2 achieves 30/30."
- **Abstract:** "HLLMNN achieves 100% recovery at all noisy conditions ($\sigma > 0$) and **90.0% (27/30 equations)** under the strict noiseless protocol ($R^2 > 0.9999$)."
- **Table 9** ($\sigma = 0\%$ row): HLLMNN recovery = **90.0%**‡, footnoted "M4 noiseless: 90% at threshold 0.999999; 100% at threshold 0.995."
- **Section 14, Conclusions, point 2:** "M3 outperforms M4 at $\sigma = 0$: 100% vs **90%** noiseless; **three further M4 noiseless failures remain** (Lorentz force, Photon energy, Zeeman energy)."

Four places in the same document agree on 90% / 27/30 with 3 named failures — and Table 4 alone says 30/30. Table 4's threshold notation (" $R^2 \geq 0.9999$ ") is the same one the abstract labels "strict" and pairs with 27/30, so the discrepancy isn't just a footnote nuance — it's a straight contradiction inside one document. Table 4 needs a correction (or a footnote making clear it's using the looser $\sigma=0.995$ -style threshold, if that's actually what happened).

● Minor — Benchmark size denominator drifts across documents (mostly explained, but worth tightening)

The main paper's Appendix B table cleanly documents version history: v1.0 = 62 cases, v2.0 = 71, v3.0 = 74 (canonical, used everywhere in the main paper). Supp. A, however, still uses a **66-case denominator** throughout its "Projected Impact on the Leaderboard" table and reproducibility commands (`--fixed-denominator 66`), and explicitly flags itself as "in-sample, not held-out" (Remark 1) — so this is disclosed, not hidden. Still, a reader moving between the main paper (74) and Supp. A (66) without reading Supp. A's Remark 1 first could easily misattribute Supp. A's 82.2%/72.7% figures to the main 74-case benchmark. Consider adding a one-line reminder at the top of every Supp. A table: "n = 66, pre-v3.0 historical — see main paper Table 2 for the v3.0/n=74 confirmed figures."

2. What I checked and found genuinely consistent (worth knowing, not just problems)

These cross-document checks held up:

- **1.73× LLM-routed speedup (n=68/74):** identical across the main paper's abstract, Contributions, §8.3, §10.4, Table 4, Table 11, Conclusion, and Supp. A §C.2 and Supp. A's reproducibility checklist.
 - **89.2% near-perfect success rate, +27.0 pp over LLM, +83.8 pp over neural net:** arithmetic checks out ($89.2 - 62.2 = 27.0$; $89.2 - 5.4 = 83.8$) and the figure is repeated identically in the abstract, Table 2, Table 3, §10.2, and Supp. A's reproducibility checklist ("confirm HypatiaX 89.2% vs LLM 62.2%").
 - **Difficulty-tier breakdown** (Easy 24 / Medium 29 / Hard 21 = 74) and the associated gains (+12.5 / +31.1 / +38.1 pp) match Table 3 exactly.
 - **Portfolio Variance seed-sweep arithmetic** (Table 5): the reported means ($P = -10.686$, $H = -6.261$) correctly average the five listed per-seed values, and "4/5 wins, 2/5 exact recoveries" matches the table's Win/Recovery columns.
 - **6 catastrophic LLM failures / 0 for HypatiaX, and Michaelis-Menten far- $R^2 \approx -635$** (abstract) vs. **-634.5989** (Table 6) — consistent.
 - **Table 1 vs. Supp. B Appendix G:** the Core-15 "1231% mean neural extrapolation error, $U=0$, $p=1.11 \times 10^{-6}$ " figures match exactly between the main paper and the independently-computed statistics in Supp. B's Appendix G.1-G.2.
 - The paper's self-corrections (73%→1.73× runtime claim; the Portfolio Variance seed=42 attribution fix; the Newton's-gravity absolute-threshold bug and its fix applied at three code locations) are each stated once and then referenced consistently wherever they recur later in the text — no version drift there.
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3. Bottom line

The paper is unusually self-critical in its prose (it flags its own bugs and corrects a previously-inflated runtime claim), which is a good sign. But two of the issues above are not cosmetic:

1. **Figures 3 & 4 are broken** and actively contradict the paper's own authoritative table (Table 6) — this will almost certainly draw a reviewer comment or a request for figure regeneration.
2. **The Section 10.7 Feynman "reduction" paragraph is internally contradictory** and needs to be rewritten with numbers that actually match the rest of the section.

The Supp. B Table 4 vs. abstract mismatch (90% vs. 100% for HLLMNN) is a smaller but real fix. The denominator drift across Supp. A is disclosed but could be signposted more clearly.

I was not able to review the cover letter, since none was uploaded to this conversation.